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Deep Learning-Based Beach Crowd Counting: Model Refinement and Multi-Camera Hardware Evaluation

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Master's Degree in Intelligent Systems (MUSI)
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***Abstract*—This work presents the design, implementation, and evaluation of a scalable and cost-effective crowd monitoring system for beach environments based on deep learning techniques for computer vision. The whole project addresses four main objectives: hardware configuration, dataset construction, crowd-counting model development and domain transfer across different hardware setups and locations.**

A multi-tier hardware setup was deployed using four different devices: an ESP-32 OV-3660, two Raspberry Pi camera modules (Camera Module 3 and Camera Module 2 NoIR) and a high-quality M12 wide-angle camera. Each device was evaluated under comparable conditions on unseen beach locations. Training images were collected from both curated public crowd-counting datasets and from publicly available webcams.

A bayesian crowd counting model, built on a VGG-19 backbone, was adopted and extended from prior work with a custom loss function, a custom metric which measures prediction error relative to each beach's maximum occupancy. Additionally, spatial prediction masks were introduced to suppress false positives in non-relevant regions such as treetops, and rooftops, yielding an accuracy improvement.

Results show that the trained model outperforms its predecessor in both MAE and MSE on the validation set. Among the tested hardware platforms, the ESP-32 OV-3660 proved to be the most cost-effective. The masking technique proved to have the greatest impact, reducing false positives by up to an order of magnitude in several cases. These findings prove that CNN-based crowd counting monitoring is practically feasible in real beach environments using affordable hardware.

I. Introduction

Monitoring crowd density in public spaces has become an increasingly important factor as it enables the prediction of high traffic, infrastructure management, improves resource allocation, and can enhance public safety when properly integrated with emergency response systems. This challenge is particularly relevant in open and dynamic environments such as beaches where crowd levels can fluctuate due to external factors like weather conditions, current season, tourism patterns or the presence of jellyfish. In such contexts, obtaining accurate and timely information about crowd distribution is essential for both operational decision-making and long-term planning.

Traditional approaches to crowd monitoring, including manual counting and classical computer vision techniques, often struggle to provide reliable results in complex real-world scenarios. Variations in lighting conditions, occlusions, perspective distortions, and high-density gatherings limit their effectiveness. However, in recent years, advances in deep learning, particularly Convolutional Neural Networks (CNNs) [4], have led to substantial improvements in crowd counting accuracy. These methods are capable of learning robust feature representations directly from data, enabling them to perform well even in challenging and unstructured environments.

Despite these advancements, several challenges remain when deploying such systems in real-world applications. Differences in camera hardware, image quality, viewpoint, and environmental conditions can greatly affect model performance. In addition, the need for scalable and cost-effective solutions is critical.

This master's thesis is part of a larger collaborative project involving multiple researchers and contributors, building upon prior work initiated in the article by Ma *et al.* [9] and subsequently extended in a final degree thesis [3]. The overarching project is structured around three main project requirements:

- predict future beach occupancy using historical occupancy data combined with additional sources such as weather conditions.
- upgrade an existing web application to incorporate enhanced functionalities for visualization and decision support.
- design, implement and validate a scalable and cost-effective crowd monitoring system based on visual data collected from public webcams, including an evaluation of suitable hardware alternatives for data acquisition.

Within this broader context, the present thesis focuses specifically on the third objective. More precisely, it addresses the refinement of computer vision models for crowd counting and the experimental evaluation of different hardware platforms capable of capturing and providing visual data to these models in real-world conditions

To achieve this, the following specific scientific objectives are defined:

1) Configure different devices for taking photographs: ESP32, Raspberry Pi, commercial cameras.
2) Collect images for crowd counting in beaches or similar sceneries.
3) Crowd-counting AI model: design appropriate metrics, compare AI models, and validate accuracy.
4) Domain transfer: fine-tune an AI model using data from one camera (or situation), and validate it with the others.

Document Structure

The remainder of this thesis is organized as follows. Section II reviews the state of the art in crowd counting, with particular emphasis on deep learning based approaches, the datasets commonly used for benchmarking, and existing approaches to domain adaptation and transfer learning relevant to deploying models that process images acquired with different camera hardware.

Section III presents the methodology followed including hardware configuration for image acquisition (Section III-A), the construction and preprocessing of the training dataset (Section III-B), the bayesian crowd counting architecture adopted (Section III-C), the evaluation metrics used to assess model performance, including the Capacity MAE, a metric introduced in this work (Section III-D), and the proposed spatial masking technique introduced to suppress false positives (Section III-E).

Section IV reports and discusses the experimental results, covering the model training process (Section IV-A) and the evaluation of different camera platforms under real-world conditions (Section IV-C).

Finally, Section V summarizes the main conclusions of this work, discusses its limitations, and outlines potential directions for future research.

II. State of the Art

Crowd density monitoring has evolved majorly over the last decade, driven by improvements in computer vision, deep learning, and edge computing. This section reviews the current landscape of crowd counting techniques, focusing on deep learning architectures, the role of hardware in scalable deployment, and strategies for domain adaptation, which are critical for the objectives of this work.

A. Evolution of Crowd Counting Techniques

Early approaches to crowd counting relied on classical computer vision techniques, such as background subtraction, blob detection, and hand-crafted feature descriptors like Histogram of Oriented Gradients combined with Support Vector Machines [1, 6]. Although they were proved to be somewhat effective in controlled environments, these methods struggled with the complexities of real-world scenarios. They were highly sensitive to variations in lighting, camera perspective, occlusions, and the non-rigid nature of crowds, often failing in high-density or dynamic settings common in public spaces [2].

B. Deep Learning Approaches for Crowd Density

The emergence of CNNs revolutionized crowd counting by shifting the paradigm from explicit detection to density regression. Modern methods typically employ a CNN to predict a density map, which can be used to yield the crowd count. This approach handles high-density crowds more effectively than detection-based methods, as it does not require distinguishing individual heads [14].

1) Density Map Regression Architectures: Key milestones in deep learning-based crowd counting include:

- **Multi-Column CNNs (MCNN):** Zhang *et al.* [14] propose an architecture that uses parallel convolutional layers with different kernel sizes to capture crowds at multiple scales, improving robustness to size variations.

- **CSRNet [7]:** combine a CNN backbone for feature extraction with dilated convolutions to expand the receptive field without increasing computational cost. This architecture has turned out to be more accurate than other models while not being too computationally expensive.

- **Transformer-based Models:** recent works integrate Vision Transformers to capture long-range dependencies in crowd distribution, such as the Vision Transformer-based approach proposed in [8] that demonstrates that attention mechanisms can improve performance in highly occluded or structured environments.

2) Datasets and Benchmarking: State-of-the-art crowd counting methods are commonly evaluated on public benchmarks such as the ShanghaiTech Dataset [14], UCF-QNRF [5], and JHU-Crowd++ [12]. These datasets cover diverse scenarios, including urban streets, subway stations, and public squares, and have become the standard benchmarks for assessing model performance. However, they provide limited representation of open, dynamic environments such as beaches, resulting in a significant domain gap when models trained on these datasets are deployed in coastal settings characterized by sparse, irregular, and highly variable crowd distributions. To address this limitation, in addition to benchmarking against these widely used state-of-the-art datasets, we also evaluate our approach on our own beach-specific dataset, which is designed to capture the unique characteristics of coastal environments and assess the model's generalization under realistic beach monitoring conditions.

C. Domain Adaptation and Transfer Learning

A critical challenge in deploying crowd counting systems is the performance degradation caused by domain shifts. Differences in camera viewpoints, lighting conditions, and background textures can have a great impact on model accuracy. Domain adaptation and transfer Learning techniques are employed to mitigate these issues:

- **Fine-Tuning:** The most common approach involves pre-training a model on a large source dataset and fine-tuning it on a small amount of target data. Recent studies suggest that fine-tuning the density head or specific CNN blocks is often sufficient for moderate domain shifts [13].
- **Generalization Strategies:** Given the cost of labeling target data, domain generalization methods are gaining attention, aiming to train models that generalize to unseen domains without target data. This is particularly relevant for the objective of validating a model trained on one camera across different beach locations or environmental conditions.

III. Methodology

This section describes the methodology followed to address the four objectives defined in Section I. First, the hardware platforms used for image acquisition are presented, followed by the dataset construction process, the crowd counting architecture, the evaluation metrics, and the proposed spatial masking technique.

A. Hardware configuration & data acquisition pipeline

To satisfy Objective 1, a multi-tier hardware setup (see Table I) was configured to balance cost, computational capability, and environmental resistance.

Table I: Specifications of the camera modules considered.

Platform	Camera Model	Max. Resolution	Approximate Cost
ESP-32	OV-3660	2048 × 1536 px	10€
Raspberry Pi	Camera Module 3 Wide	4608 × 2592 px	30€
Raspberry Pi	Camera Module 2 NoIR	3280 × 2464 px	30€
Raspberry Pi	High Quality Camera M12 + wide-angle lens 185°	4056 × 3040 px	110€

All devices were configured to send the captured images to a public web server and also to save them locally (in their internal persistent memory) for further analysis in case the connection is unstable.

B. Dataset construction, acquisition & preprocessing

Training a deep learning model requires a substantial number of labeled images. In particular, the training model should consist of images from the same domain and exhibiting characteristics representative of those expected during deployment thereby enabling the model to generalize effectively to real-world operating conditions.

The final dataset incorporates images collected and generated during earlier stages of the project (June 2022). These images were obtained from publicly accessible webcams, which are typically owned by hotels or internet service providers. While many of the collected images did not provide useful information for the intended tasks, a subset proved to be suitable for inclusion in the dataset. Because these images were not originally annotated for supervised learning, they underwent a manual review and labeling process before being incorporated into the final dataset.

In addition to manually labeled data, several publicly available datasets were downloaded [10] and combined to increase the quantity of training samples. These datasets provided a broad range of examples captured under different lighting conditions and camera viewpoints. To reduce the domain gap with our target application, a subset of the public images was manually selected.

Labeling process: All manually annotated images followed the same labeling criterion. Because often times individuals at small scales, or partially occluded, or blurry caused confusion. To address this, we adopted a confidence-based guideline to ensure consistency across ambiguous cases. The annotator was instructed only to place a label whenever they were more than 50% confident that those pixels indeed represented a human.

My contribution to this process was to define the image selection criteria and determine which samples were retained, rather than performing the annotation itself. Specifically, only images depicting beach environments from a low-oblique, near-horizontal viewing perspective (i.e., cameras with optical axes approximately parallel to the ground, rather than nadir or steep aerial viewpoints) were included. Figure 1 illustrates the original diversity of the public datasets alongside the filtered subset, highlighting the retained images that best match the viewpoint and scene characteristics of our own dataset.

C. Bayesian crowd counting model architecture

The bayesian crowd counting approach proposed in [9] was selected as the crowd estimation model due to its performance and ability to generalize across different environments and viewpoints.

The implementation employs a VGG19-based [11] convolutional neural network as the feature extraction backbone. Fully connected layers originally present in VGG-19 for image classification are removed.

After feature extraction, the network uses a density regression head that transforms the extracted features into a continuous density map. The final output is a density representation where each pixel value contributes to the final crowd estimation. The total estimated crowd count is obtained by integrating the predicted density map over the

Figure 1: Dataset sample images. Public dataset (top). Self-collected image from Son Serra (bottom).

image domain.

D. Evaluation metrics

Conventional evaluation metrics such as mean absolute error (MAE) or mean squared error (MSE) provide valuable information regarding model performance. However, these metrics may not always convey the practical significance of the prediction error. For instance, a MAE of 20 is clearly preferable to a MAE of 200, but the magnitude of a 20-person error cannot be interpreted without considering the context in which it occurs.

As illustrated in Figure 2, an absolute error of 20 units has different implications depending on the characteristics of the beach. Specifically, committing an absolute error of 20 units on the beach shown on the top represents a bigger problem than the same error on the beach at the bottom, as it has a greater real-world surface; hence, the quantity of people that can be in that image (capacity) is also greater. Therefore, normalizing the error by beach capacity provides a more meaningful metric.

To address this limitation, we introduce a new metric, **Capacity MAE**, which was not considered in prior work [3]. Rather than measuring the absolute error in the predicted count, Capacity MAE quantifies the prediction error relative to the capacity of each beach and computes the mean value across all samples. An illustrative example of its calculation is presented in Table II.

Table II: Illustrative example of the Capacity MAE metric. Although the absolute prediction error is identical for all four samples (20 people), the Capacity MAE varies according to the beach capacity, providing a normalized measure of prediction error across beaches of different sizes.

Beach Name	True Count	Capacity	Prediction	Absolute Error	Capacity MAE
A	75	115	95	20	17.39%
B	740	935	720	20	2.13%
C	2	12	22	20	166.67%
D	15	40	35	20	50.00%

Figure 2: Different beaches can have different capacities.

Our objective is not to predict the *exact* crowd count for each specific beach, but rather to provide a rough estimate and try to make it as close to reality as possible for each beach's capacity. That should correspond to making the value for Capacity MAE as low as possible. To guide the model towards this objective, we can modify its loss function.

The model's loss function implemented by default is bayesian loss which follows Eq. 1

$$L = \frac{1}{B}\sum_{b=1}^{B}\sum_{i=1}^{N_b}\left|t_i^{(b)} - \sum_{j=1}^{HW} D_j^{(b)} P_{ij}^{(b)}\right| \tag{1}$$

Where

- B: batch size
- N_b: number of annotations for image b
- $t_i^{(b)}$: target count for annotation i (typically 1)
- $D_j^{(b)}$: predicted density at pixel j

- $P_{ij}^{(b)}$: probability that the pixel j belongs to the annotation i
- HW: total number of pixels

Essentially, it computes an L1 absolute error between the real count and the predicted count computed by the density map and then taking the average. We have added a new term to this loss function so that it also takes into account the capacity error. See Eq. 2.

$$L = (1-\lambda)\frac{1}{B}\sum_{b=1}^{B}\sum_{i=1}^{N_b}\left|t_i^{(b)} - \sum_{j=1}^{HW} D_j^{(b)} P_{ij}^{(b)}\right| + \lambda\frac{1}{B}\sum_{b=1}^{B}\left|100\cdot\frac{C_b}{M_b} - \hat{K}_b\right| \tag{2}$$

Where:

- C_b: true crowd count for image b
- M_b: maximum beach capacity
- $\hat{K}_b$: predicted beach occupancy ratio
- λ: capacity weight [0, 1]

The Capacity MAE term is weighted by the hyperparameter λ which controls the contribution of the Capacity MAE to the overall loss.

E. Prediction masks

It was observed that the model would often mistake image areas as large clots of people, especially when lighting conditions changed and created shadows. Figure 3 shows two clear examples of this issue being one of them (Figure 3, bottom) especially grave since the model is predicting a total count of 398 (most of them allegedly on the trees) but the beach is actually empty. These false positives can dominate the final prediction making it unsuitable for final deployment.

To mitigate this issue, we applied a masking strategy that excludes irrelevant regions of the images from consideration. Specifically we restrict the analysis to areas where people are likely to appear, ensuring that predictions outside these regions are disregarded and do not contribute to the final result.

As described in Section III-C, the model outputs a density map over the input image, the sum of which yields the total estimated crowd count. To mask out unwanted predictions we calculated the element-wise product (Hadamard product) between the model's output and our mask. So the new pipeline, illustrated in Figure 4 is:

1) get the snapshot,
2) calculate crowd count heatmap (`H`),
3) calculate `Output` $=$ `H` $\odot$ `Mask`,
4) calculate `count` $= \sum_{i,j}$ `Output`$_{i,j}$,

The masking stage is applied after the CNN generates the density map. This avoids modifying the network itself.

Figure 3: Examples of model failing. The model estimates 265 people on the image on the top and 398 people on the image on the bottom.

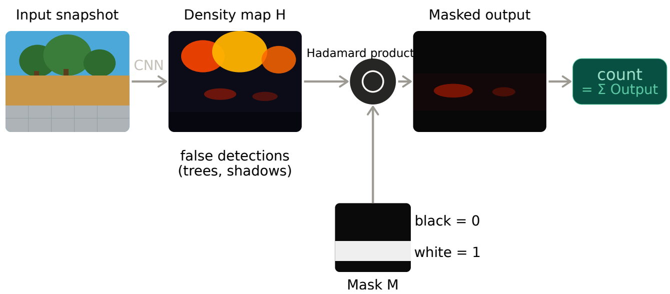

Figure 4: Modified crowd counting pipeline incorporating spatial masks. The density map output is element-wise multiplied with a binary mask to exclude irrelevant regions prior to computing the final count.

F. Model training

The bayesian crowd counting model described in Section III-C was trained using a mixture of self-collected beach images and selected samples from publicly available datasets with different point of view angles, lighting, crowd density and image quality. Table III displays the exact counts for the train and validation splits.

As specified by the main project's implementation [9] [1], all images and their annotations had to undergo a preprocessing procedure prior to training. This does not alter the data in

[1] Z. Ma. Bayesian-Crowd-Counting. https://github.com/zhiheng-ma/Bayesian-Crowd-Counting, 2020. GitHub repository.

Figure 5: Top image is the mask, bottom image is results after mask was applied.

Table III: Train & Validation split counts.

	Train	validation
Self-collected	181	72
Public Dataset	5970	0

any form but rather makes it so that it can be fed to the model.

Regarding hyperparameters, λ (see Eq. 2) was set to 0.6, while all other hyperparameters were set to their default values.

IV. Results & Analysis

This section reports the experimental results in two parts: the training and validation of the extended bayesian crowd counting model, and the comparative evaluation of the four camera platforms on previously unseen beaches.

A. Model training

The final model was trained on a machine with an NVIDIA A6000 GPU with 48GB of VRAM and 128 GB of RAM for a total of 1000 epochs or 3 hours. Table IV summarizes the final results and compares them to the results obtained before this work. Figure 6 displays how these values evolved over the entire training.

Table IV: Validation set metrics for the previous model and the new model. The old model is the existing model prior to this work, developed by J. F. Sánchez Garcia in [3]. The new model is the final iteration of the model presented in this work.

	Old Model	New Model	Comparison
MAE	28.70	6.38	77.8% decrease
MSE	37.01	9.39	74.6% decrease
Capacity MAE	No data	41.02%	N/A

As shown in Figure 6, the Capacity MAE curve displays an alarming behavior when compared to the MAE and MSE curves. While the training Capacity MAE converges rapidly to low values, the validation Capacity MAE remains persistently high at around 41% throughout the entire training process, showing no sign of improvement. An analysis was conducted to identify the cause of this discrepancy. The results revealed that, since the Capacity MAE is defined as an error relative to each beach's capacity, it is highly sensitive to outliers: beaches with very small capacities can produce disproportionately large relative errors even when the absolute counting error is modest. For instance, as previously illustrated in Table II, an absolute error of 20 people on a beach with a capacity of 12 yields a Capacity MAE of 166.67%, whereas the same absolute error on a large beach contributes only marginally. A small number of such low-capacity samples in the validation set is therefore sufficient to dominate the mean and inflate the aggregate metric, masking the fact that the model performs adequately on the majority of beaches.

B. Masks impact

After applying this technique, many webcams that were providing highly inaccurate estimates, started to provide closer-to-reality results. Figure 5 shows the results after applying the mask on the same image analyzed previously (see Figure 3). Before masking, the model predicted 265 people (most of them on rooftops or treetops). After masking, the prediction decreased to 31, compared with the true count of 3. Although the prediction remains above the true count, the reduction demonstrates that most of the original error originated from irrelevant image regions rather than the beach itself thus confirming that spatial masking effectively suppresses systematic false positives.

C. Performance on new beaches

To evaluate the generalization capability of the trained model and test which of the four cameras used (see Table I) provided a better result. These cameras have been tested under the same conditions, that is, the same viewpoint, lighting environment, and crowd density. Moreover, snapshots taken belong to a beach that the model had not seen before. Table V summarizes the results obtained.

The results reveal notable differences in model accuracy across camera hardware, which will need to be further broken

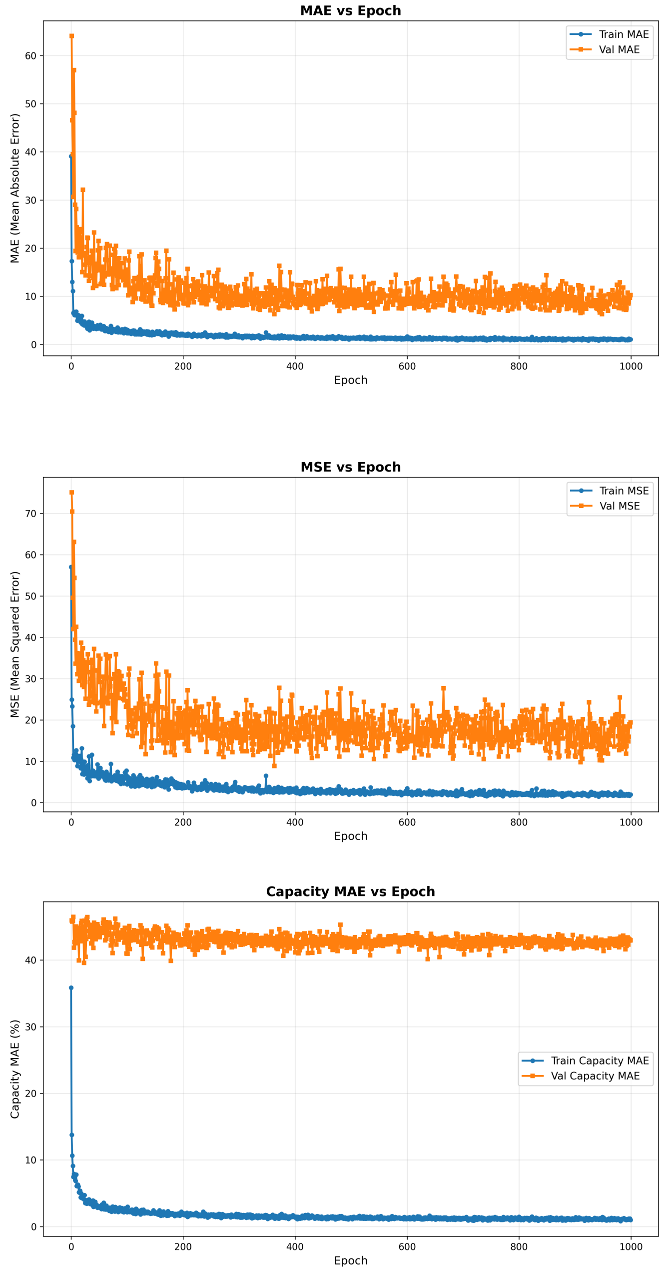

Figure 6: MAE, MSE, and Capacity MAE curves for model training.

down.

Firstly, *Camera Module 2 NoIR* gave the worst performance. This was already expected since its lack of an infrared filter alters the apparent color and texture of the scene. Figure 7 illustrates the appearance of images captured by the Raspberry Pi Camera Module 2 NoIR. Due to the absence of an infrared filter, the captured images exhibit a noticeable pinkish tint and altered texture compared to standard RGB images. These visual differences represent a domain shift with respect to the training data, which consisted exclusively of visible-spectrum images. As a result, the model struggles to extract the same visual features it learned during training, leading to higher prediction errors than the other camera modules.

Table V: Model performance over different camera models.

Camera Model	Average True Count	MAE	Total Snapshots
OV-3660	32	19	5
RPi Camera Module 3 Wide	56	51	4
RPi Camera Module 2 NoIR	47	87	3
RPi High Quality Camera M12 + wide-angle lens 185°	71	43	3

Figure 7: Prediction from a snapshot taken with camera module 2 NoIR.

The OV-3660 sensor achieved the lowest MAE, which was somewhat surprising given that it is the lowest-performing sensor overall. However, this result should be interpreted with caution. The snapshots captured by the OV-3660 sensor contained the fewest people, reducing the potential for counting errors. Additionally, the images were acquired at a sufficiently close range, minimizing the impact of the sensor's limited capabilities. Despite these factors, the OV-3660 sensor, also produced the fewest false positives. In fact, it generated no false positives. Figure 8 presents the density heatmap for a representative snapshot captured with the OV-3660 sensor. In contrast to the heatmaps produced with the other sensors, no regions without people exhibit elevated density values, indicating the absence of false positive activations.

Although *Camera Module 3 Wide* and the High Quality Camera M12 have a similar MAE, one could argue that M12 performed better because there are more people in snapshots taken by it. However, if we take a closer look at the generated heatmaps (see Figure 9), we can see that the predictions on the M12's snapshots contain more false positives in areas where there is clearly no one there such as close-by sand areas where image texture can be more dense. In contrast, the false positives observed in snapshots captured by the Camera Module 3 Wide sensor tend to occur in regions containing

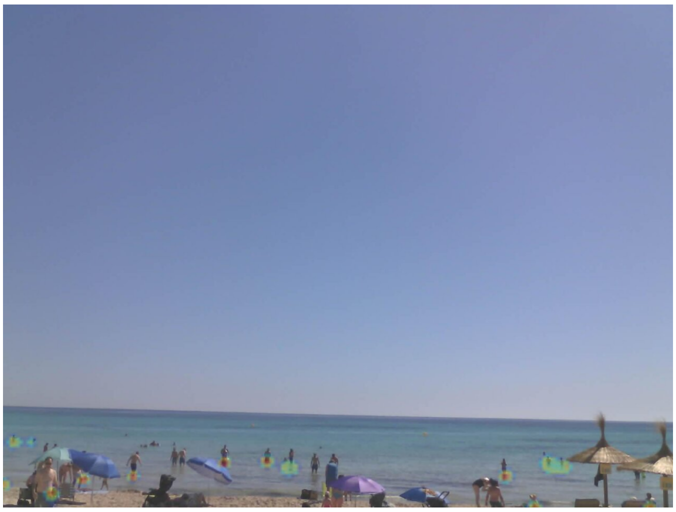

Figure 8: Heatmap of a snapshot taken by OV-3660.

objects or visual features that are ambiguous and may resemble human subjects such as beach balls, buoys, or areas beneath parasols.

V. Conclusions

This work presented the design, implementation and evaluation of a scalable and cost-effective crowd monitoring system addressing four specific objectives (see Section I). Despite being centered in beach environments, it would not be difficult to change it to a different setting (urban, rural, etc.). Only labeled images and a model retrain would be necessary.

Regarding hardware configuration, data acquisition and domain transfer (Objectives 1 and 4), a multi-tier setup incorporating four different sensors was successfully deployed. The evaluation yielded both expected and unexpected findings. As anticipated, the NoIR sensor exhibited the poorest performance. Conversely, the OV-3660 sensor demonstrated a remarkably cost-effective performance despite its hardware limitations, while the M12 sensor delivered lower than expected accuracy, likely due to the limited representation of images with such wide field of view in the training dataset.

In terms of data collection (Objective 2), a heterogeneous dataset was assembled combining self-collected beach images and publicly available crowd counting datasets. However, images belonging to different scenarios such as roads, urban spaces, etc. were discarded since they did not belong to our domain.

With respect to the crowd counting AI model (Objective 3), the bayesian crowd counting architecture was adapted and extended with a custom loss function that incorporates a Capacity MAE term. The trained model achieved better results on the validation set than the previous model in terms of MAE and MSE. However, the feature with the greatest

Figure 9: Heatmap of snapshots taken by different sensors. Top image is High Quality Camera M12 + wide angle lens 185º. Bottom image is Camera Module 3 Wide.

impact was the introduction of spatial prediction masks, which reduced the predicted count by one order of magnitude for several samples.

Although there have been some major improvements to the previous project, some limitations still remain:

- The Capacity MAE remains high at 41.02% indicating that absolute beach occupancy still requires improvements, since occupancies were calculated by extracting the maximum value for each beach from labeled data.
- The masking technique, while really effective, is only applicable to fixed-viewpoint cameras. If a camera is repositioned or its field of view changes, the mask must be manually updated.
- The tests performed with snapshots of beaches that were also included in the training set yielded higher accuracy results. This suggests that incorporating representative training data for every beach would substantially reduce the model's predicting error. However, the effort required to manually label 10–20 snapshots for every beach represents a burden.
- The capacity for every beach was calculated as the

maximum count of all the images labeled for that beach. However, this may be the most suitable method since we had a limited number of images for each beach.

Overall, this work demonstrates that CNN-based crowd monitoring is feasible in real environments, especially beaches, using affordable hardware.

REFERENCES

[1] A. B. Chan, Z.-S. J. Liang, and N. Vasconcelos. Privacy preserving crowd monitoring: Counting people without people models or tracking. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1–7, 2008.

[2] Z. Fan, H. Zhang, Z. Zhang, and G. Lu. A survey of crowd counting and density estimation based on convolutional neural network. *Neurocomputing*, 2021.

[3] J. F. S. García. Monitoreo automático de la ocupación de las playas de las islas baleares, 2024. Escuela Politécnica Superior, curso 2023–2024.

[4] I. Goodfellow, Y. Bengio, and A. Courville. *Deep Learning*. MIT Press, Cambridge, MA, USA, 2016.

[5] H. Idrees, M. Tayyab, K. Athrey, D. Zhang, S. Al-Maadeed, N. Rajpoot, and M. Shah. Composition loss for counting, density map estimation and localization in dense crowds. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 532–546, 2018.

[6] V. Lempitsky and A. Zisserman. Learning to count objects in images. In *Advances in Neural Information Processing Systems*, volume 23, pages 1324–1332, 2010.

[7] Y. Li, X. Zhang, and D. Chen. Csrnet: Dilated convolutional neural networks for understanding the highly congested scenes. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1091–1100, 2018.

[8] D. Liang, W. Xu, and X. Bai. An end-to-end transformer model for crowd localization. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 38–54, 2022.

[9] Z. Ma, X. Wei, X. Hong, and Y. Gong. Bayesian loss for crowd count estimation with point supervision. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 6141–6150, 2019.

[10] Roboflow. Roboflow Universe: Free Computer Vision Datasets and Models. https://universe.roboflow.com/, 2026. Accessed: 2026-07-02.

[11] K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.

[12] V. A. Sindagi, R. Yasarla, and V. M. Patel. Pushing the frontiers of unconstrained crowd counting: New dataset and benchmark method. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1221–1231, 2019.

[13] Q. Wang, J. Gao, W. Lin, and Y. Yuan. Learning from synthetic data for crowd counting in the wild. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[14] Y. Zhang, D. Zhou, S. Chen, S. Gao, and Y. Ma. Single-image crowd counting via multi-column convolutional neural network. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 589–597, 2016.